

Course Code: BISM2207

Course Title: Managing Business Data

Assessment 2: System Development
FINAL REPORT



**THE UNIVERSITY
OF QUEENSLAND**
A U S T R A L I A

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Green Purchasing Management System

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Project Proposal:

The Olympics is a global sport competition and attracts a large audience every four years. However, it shows that the flow of carbon emission is diminishing every time, so it is really important for the Olympics in Brisbane to try to reduce the carbon emissions. The Green Purchasing Management System focuses on responsible consumption and production (SDG 12) and climate action (SDG 13). The system's purpose is to manage and reduce the environmental impact of the Brisbane 2032 Olympic Games through sustainable practices in purchasing, waste management, and resource allocation. The system will collect and process data related to the carbon footprint and reusability of products used in the event. By doing so, the system can minimise the environmental impact of the event and align with global sustainability goals. Furthermore, by integrating data from various sources, including manual entries and external databases, the system provides comprehensive insights into the sustainability performance of products and services and the application uses this data to make comparison. This approach promotes collaboration with companies that share a commitment to reducing their environmental impact. By implementing our proposed system, there will be a lot of benefits. First of all, it will help to increase the efficiency of reducing the carbon footprint during purchasing and in the venue, which can help the Olympics in Australia to meet the SDG12 and SDG13 and even other industries with other problems. Secondly, it can make the improvement more easier by collecting data, we can easily know the result by input different data in it. For instance, we can enter product's data and other entities' data to find out which product can be improved or replaced for the reduction of carbon footprint, instead of spending a lot of time to calculate each by yourself. What's more, after setting up a rating system, we will have a higher efficiency to help improve and grade the carbon footprint emission of each product or sport. Next, the solid and scalable system can be used in different organisations. For instance, different contests or activities can use our system if we conclude different aspects that they need to pay attention to and can make our system a long term system for their development. Last but not least, if we use the system in the Olympics or other activities, it can show that we do care about the environment and are willing to take action or improve it. By doing so, it will help to improve the brand image and gain customers preference.

Narrative of the Functions (the management system) the applications will perform:

INPUT:

Firstly, There will be a lot of input data being stored in the system. We need to find ways to collect different data. For instance, the venue and sport will be important and we can presume it from the past information and we can also identify the amount of infrastructure after finding the venue and sport. For the number of participants involved in the sport, we can use Questionnaire to estimate the number. We can get each participant's full name and date of birth by including in the questionnaire too. Also, we can find product information(quantity of the production, can it be reused, amount of carbon emissions caused by products) and service information by asking different suppliers for their past record. Collaboration with different suppliers can also help them improve their performance.

PROCESSING:

The data-organising processes within our system cover data extraction from numerous external databases and manual data entries. The data extraction function extracts data from external databases, including detailed reports of (1) reusability and (2) carbon footprint emission for each product discarded. Manual data entry includes Brisbane 2032 Olympic Games' records of (1) sports held, (2) participants in each sport, (3) match schedules, (4) product containers available and sold / given in each match, and (5) list of all suppliers / producers for each product.

Our system will organise, separate, and ensure clean data input for manual data entries. Several fundamental instances would be the development of (1) the “*Service*” table, which tracks how a certain service performed in a venue would count for products discarded; and (2) the “*ProductVenue*” table, which tracks which products are discarded the most in each venue. These two tables are important for system analysis. Additionally, the system ensures that only distinct / unique data will be entered here.

Data from external databases shall include carbon footprint emissions (in kgCO₂), which are used to comprehend most harmful products for the environment. The reusability status and carbon footprint rating scores of all products are also listed to identify the ones contributing to long-term sustainability. These databases would serve as input data for our system to filter and monitor the statistics of each product. The system then filters suppliers to prioritise companies committed to sustainability based on various criterias; such as past environmental performances, adherence to sustainability in previous events, history of actions taken to reduce carbon footprint, etc. In the final steps, the system would integrate manual data entries and external databases to generate (1) comparisons of product & supplier combinations through the rating system; and (2) sets of options / actions aiming to reduce carbon emission.

Output

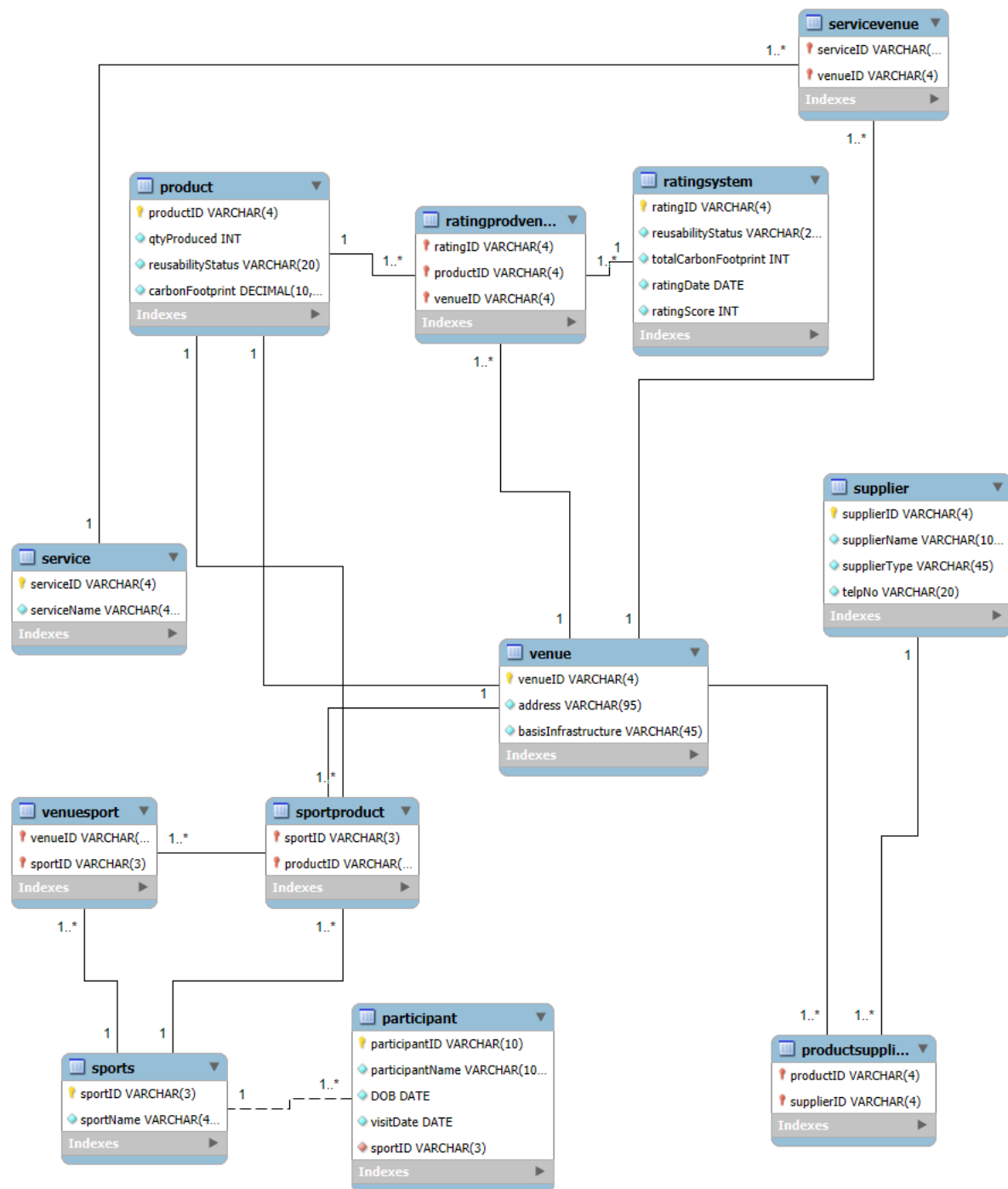
The **output** of the system is a series of reports that provide insights into the event's sustainability efforts and resource management. These reports are generated automatically from the processed data and help event organisers make informed decisions to improve sustainability. The major reports include:

- 1. Product Options Report:** This report shows the average carbon footprint and average user rating score for each product. It helps the event organisers identify

products that are both high-quality and environmentally friendly, supporting future procurement decisions.

2. **Top 3 Carbon Footprint Report:** This report identifies the top three products that have the highest carbon footprint during a specified period of the event. It helps organisers focus on reducing or replacing these high-impact products to lower overall emissions.
3. **Supplier Evaluation Standards Report:** This report evaluates the performance of suppliers based on the total quantity of products they provided and the percentage of those products that are recyclable. Suppliers with higher recycling percentages are highlighted as more sustainable partners for future events.
4. **Total Carbon Emission for Each Service Report:** This report provides a breakdown of the total carbon emissions associated with each service (e.g., catering, transport) provided during the event. By identifying high-emission services, organisers can target those areas for improvement and find ways to reduce their environmental impact.
5. **Top 3 Sports by Carbon Emission Report:** This report ranks the top three sports that generate the most carbon emissions based on the products used in those events. It helps event organisers identify which sports have the highest environmental impact and consider changes to reduce emissions in the future.
6. **Top 5 Reusable Product Performance Report:** This report highlights the top five reusable products that have received the highest performance ratings from users. These products are not only environmentally sustainable but also perform well in practice, making them ideal choices for future events.

Entity-relationship diagrams:



Data Dictionary:

Underlined = primary key

Italic underlined = foreign key

TABLE NAME:	SPORTS							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>sportID</u>	VARCHAR(3)	character: size 3, range S01-S99	NN	Y	✓		A unique identifier for each sport	Sport identification
sportName	VARCHAR(45)	character: maximum size 45	NN	N			The name of the sport	Sport identification

TABLE NAME:	SPORTPRODUCT							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>sportID</u>	VARCHAR(3)	character: size 3, range S01-S99	NN	Y	✓		A unique identifier for each sport	Sport identification
<u>productID</u>	VARCHAR(4)	character: size 4, range P01-P999	NN	N		✓	A unique identifier for each product	Product identification

TABLE NAME:	VENUE							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>venueID</u>	VARCHAR(4)	character: size 4, range ST01-ST99 / AR01-AR99	NN	Y	✓		A unique identifier for each venue (stadium or arena)	Venue identification
address	VARCHAR(95)	character: Store Venue Area Name	NN	N			If the venue is within a larger complex, you might want to	Venue identification

							store the name of the park or area.	
basisInfrastructure	VARCHAR(45)	character	NN	N			Identify the amount of infrastructure	Infrastructure identification

TABLE NAME:	VENUESPORT							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>venueID</u>	VARCHAR(4)	character: size 4, range ST01-ST99 /AR01-AR99	NN	Y	✓		A unique identifier for each venue (stadium or arena)	Venue identification
<u>sportID</u>	VARCHAR(3)	character: size 3, range S01-S99	NN	N		✓	A unique identifier for each sport	Sport identification

TABLE NAME:	PRODUCT							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>productID</u>	VARCHAR(4)	character: size 4, range P01-P999	NN	Y	✓		A unique identifier for each product	Product identification
qtyProduced	INT	Non-negative integers	NN	N			The quantity of the production	Record the number of units produced for each product
reusabilityStatus	VARCHAR(20)	character: size 20	NN	N			Whether it can be reused	Indicate the sustainability or environmental impact
carbonFootprint	DECIMAL(10,2)	numerical value and two decimal places (kgCO ₂)	NN	N			The amount of carbon emissions caused by products	Measure and record the environmental impact of each product in terms of the carbon emissions.

TABLE NAME:	PRODUCTsupplier							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>productID</u>	VARCHAR(4)	character: size 4, range P01-P999	NN	Y	✓		A unique identifier for each product	Product identification
<u>supplierID</u>	VARCHAR(4)	character: size 4, range SP01-SP99	NN	N		✓	A unique identifier for each supplier	Supplier Identification

TABLE NAME:	SUPPLIER							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>supplierID</u>	VARCHAR(4)	character: size 4, range SP01-SP99	NN	Y	✓		A unique identifier for each supplier	Supplier identification
supplierName	VARCHAR(100)	character: size 100	NN	N			The full name of the supplier	Supplier identification
supplierType	VARCHAR(45)	character: size 45	NN	N			A classification of the method of suppliership provided	Supplier classification
telpNo	VARCHAR(20)	character: maximum size 20, including international codes, area codes, various formats and separators	NN	N			The telephone number of the supplier	Supplier identification

TABLE NAME:	PARTICIPANT							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>participantID</u>	VARCHAR(10)	Character: size 10: P00000 0001-P999999 999	NN	Y	✓		This attribute stores a unique code that identifies each participant.	Participant identification

participantName	VARCHAR(100)	Any valid string	NN	N			This attribute stores the participant's full name.	Participant identification
DOB	DATE	Date (YYYY-MM-DD)	NN	N			This attribute stores the participant's date of birth.	Participant identification
visitDate	DATE	Date (YYYY-MM-DD)	NN	N			This attribute stores the participant's visiting date	Participant identification
<u>sportID</u>	VARCHAR(3)	character: size 3, range S01-S99	NN	N		✓	This attribute stores the ID of the sport associated with the participant.	Sport identification

TABLE NAME:	SERVICE							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>serviceID</u>	VARCHAR(4)	character: size 4, range SE01-SE99	NN	Y	✓		A unique identifier for each service	Service identification
serviceName	VARCHAR(100)	character: size 100	NN	N			The full name of the service	Service identification

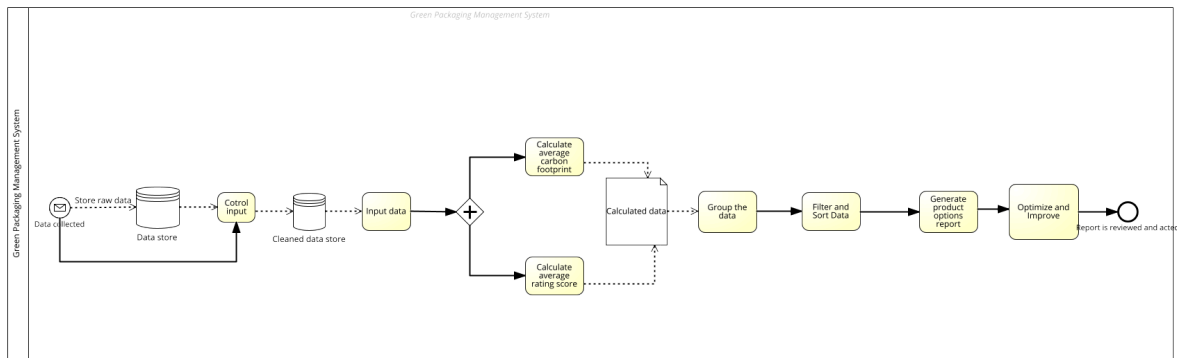
TABLE NAME:	SERVICEVENUE							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>serviceID</u>	VARCHAR(4)	character: size 4, range SE01-SE99	NN	Y	✓		A unique identifier for each service	Service identification
<u>venueID</u>	VARCHAR(4)	character: size 4, range ST01-ST99/AR01-AR99	NN	N		✓	A unique identifier for each venue (stadium or arena)	Venue identification

TABLE NAME:	RATINGSYSTEM							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>ratingID</u>	VARCHAR(4)	character: size 4, range R01-R999	NN	Y	✓		A unique identifier for each rating mission	Service/Product Rating identification
reusabilityStatus	VARCHAR(20)	character: size 45	NN	N			Whether it can be reused	Indicate the sustainability or environmental impact
totalCarbonFootprint	INT	Non-negative integers	NN	N			The number of total carbon footprint	A combination of carbon emission, transportation and usage
ratingDate	DATE	Date (YYYY-MM-DD)	NN	N			The date of rating of product/service	Showing which date is going to rate for the product or service
ratingScore	INT	Non-negative integers	NN	N			The number that identified the performance of the product and service	Showing the performance of the product/service

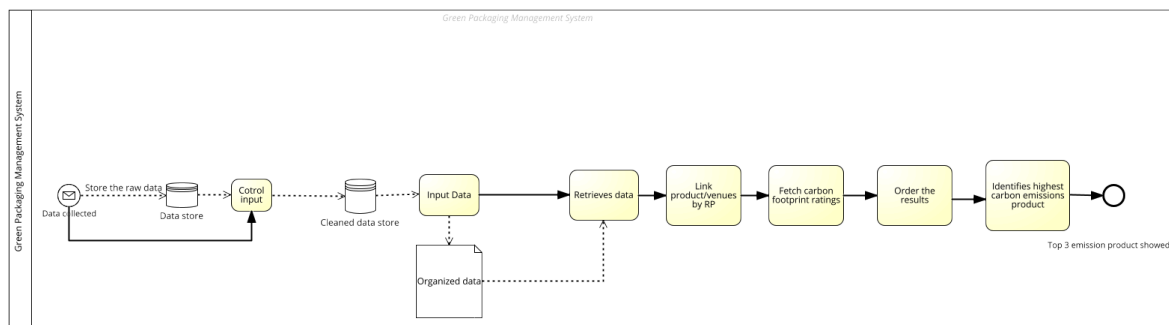
TABLE NAME:	RATINGPRODVENUE							
Attribute Name	Data Type	Domain	Null / Not Null	Uniqueness	Primary Key	Foreign Key	Definition	Purpose
<u>ratingID</u>	VARCHAR(4)	character: size 4, range R01-R999	NN	Y	✓		A unique identifier for each rating mission	Service/Product Rating identification
<u>productID</u>	VARCHAR(4)	character: size 4, range P01-P999	NN	N	✓		A unique identifier for each product	Product identification
<u>venueID</u>	VARCHAR(4)	character: size 4, range ST01-ST99/AR01-AR99	NN	N	✓		A unique identifier for each venue (stadium or arena)	Venue identification

Flowcharts or process model

Product options report: Collect data from different parties and methods or extract data from external databases - Store the raw data in a staging database before any processing takes place - Input control (Our system will organize, separate, and ensure clean data input for manual data entries.) - After validation, move the cleaned data into a structured database or data warehouse (such as SQL or cloud-based storage) for long-term storage and retrieval. - Input data - System calculates average carbon footprint - System calculate average rating score - Group the data by product ID and supplier ID, ensuring the output provides distinct product-supplier pairs - Filter and Sort Data(Order the results based on the average rating score in ascending order and also set filters based on thresholds for carbon footprint or rating score) - Generate product options report (Shows each product's sustainability insight) - Optimization and Continuous Improvement - Report is reviewed and acted upon



Carbon Footprint report during event: Collect data from different parties and methods or extract data from external databases -Data Storage and Organization:(Data Staging Area: Store the raw event-related and product-related data in a staging area for initial processing.- Input control - Move validated data into a structured data warehouse, where tables are organized by categories such as events, products, venues, and carbon footprint ratings) - Input data - The query retrieves data to track carbon footprint data - Uses the ratingprodvenue table (RP) to link products with venues - rating system to fetch carbon footprint ratings for each product at each venue - The query orders the results by the carbon footprint values in descending order – System identifies the products with the highest carbon emissions during the event - Show the top 3 highest emission products during events



Input controls

System Control

(1) Normalization (Insertion, deletion and modification test)

UNF to 1NF: To reach 1NF, we want each row to be uniquely identifiable, so we assign a unique ID to each table. For example, we add a unique ID to supplierName, which is supplierID, and attach it to the Product table as a foreign key. In this way, when the supplier experienced a corporate restructuring or merge and its name was changed, we just need to update the new supplierName in the Supplier table so that we can ensure all related information is immediately consistent.

1NF to 2NF: Assume that we have a SportProduct table={sportID, productID, sportName, noParticipant, qtyProduced, reusabilityStatus, carbonFootprint}, indicating that each sport is associated with a specific product. However, sportName and noParticipant depend only on sportID, not on productID, which means the non-key attributes are not functionally dependent on the entire primary key. To reach 2NF, we separate the product information into its own table: Product={productID, qtyProduced, reusabilityStatus, carbonFootprint}, Sport={sportID, sportName, noParticipant, productID}. In this way, you can safely delete the records related to a specific sport (e.g. platform diving) from the Sport table without accidentally deleting the product information (e.g. diving board) in the Product table. This ensures that data for other events using the same diving board remains intact.

2NF to 3NF: Assume we have a Productsupplier table={productID, productName, qtyProduced, supplierID, supplierName, supplierType}. In this table, a productID is associated with a supplierID, and each supplierID has a corresponding supplierType, which means that supplierType indirectly depends on productID through supplierID. To reach 3NF, we need to eliminate this transitive dependency by separating product information and supplier information into two tables: Product table={productID, productName, qtyProduced, supplierID}, supplier table={supplierID, supplierName, supplierType}. With this normalization, we can now insert product records independently without having to provide supplier information each time. The supplierID can be referenced as a foreign key, and if there is no supplier information available, the supplierID field can be left blank, avoiding insertion anomalies. This solution not only removes redundancy but also ensures data integrity.

(2) Integrity Constraint

Our database fulfils four types of integrity controls (Connolly, 2015; GeeksForGeeks, 2024):

a. Required-Data Constraint / NOT-NULL Constraint

All the tables within our database have fields/columns that are specified as **NOT NULL**. This ensures that no data will be missing in any row within any table. For example, the table **service** contains two fields: the non-NULL primary key **serviceID**, and the **serviceName** field specified as NOT NULL.

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`service` (  
  `serviceID` VARCHAR(4) NOT NULL,  
  `serviceName` VARCHAR(45) NOT NULL,  
  PRIMARY KEY (`serviceID`),  
  UNIQUE INDEX `serviceID_UNIQUE` (`serviceID` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

b. Domain Constraint

Wherever possible, the domain constraint is specified in the MySQL DDL using the CHECK statement. For example, the table **product** specifies that the carbon footprint level in the field **carbonFootprint** should not have a negative value.

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`product` (  
  `productID` VARCHAR(4) NOT NULL,  
  `qtyProduced` INT NOT NULL,  
  `reusabilityStatus` VARCHAR(20) NOT NULL,  
  `carbonFootprint` DECIMAL(10,2) NOT NULL,  
  CHECK (carbonFootprint >= 0),  
  PRIMARY KEY (`productID`),  
  UNIQUE INDEX `productID_UNIQUE` (`productID` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

For more complicated constraints which may not be easily accommodated in MySQL DDL statements, constraints shall be enforced in the business logic at the application level. For example, the application shall enforce that the **reusabilityStatus** should not contain spaces only. In the absence of such check at the application level, the **reusabilityStatus** field containing only spaces, such as “ ”, will be taken as correct and accepted by the DBMS.

c. Entity Integrity

The entity integrity constraint is enforced through the primary key. For example, the **sports** table contains the primary key **sportID VARCHAR (3)**.

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`sports` (  
  `sportID` VARCHAR(3) NOT NULL,  
  `sportName` VARCHAR(45) NOT NULL,  
  PRIMARY KEY (`sportID`),  
  UNIQUE INDEX `sportID_UNIQUE` (`sportID` ASC) VISIBLE)  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

d. Referential Integrity

This refers to the foreign key. A foreign key must contain a value that must refer to an existing, valid row in a parent table. Take, for example, the **participant** table, which contains **sportID** as a foreign key, referencing the parent table **sport**. Here, we assume that each participant may only participate in a single sport.

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`participant` (  
  `participantID` VARCHAR(10) NOT NULL,  
  `participantName` VARCHAR(100) NOT NULL,  
  `DOB` DATE NOT NULL,  
  `visitDate` DATE NOT NULL,  
  `sportID` VARCHAR(3) NOT NULL,  
  PRIMARY KEY (`participantID`),  
  UNIQUE INDEX `participantID_UNIQUE` (`participantID` ASC) VISIBLE,  
  INDEX `participant_sportID_idx` (`sportID` ASC) VISIBLE,  
  CONSTRAINT `participant_sportID`  
    FOREIGN KEY (`sportID`)  
    REFERENCES `greenpackaging`.`sports` (`sportID`))  
ENGINE = InnoDB  
DEFAULT CHARACTER SET = utf8mb4  
COLLATE = utf8mb4_0900_ai_ci;
```

Human Control

1. Sensitive Data Validation

Some data fields contain sensitive information that requires special attention because automated systems may not fully validate them. A common example is personal identification numbers or financial data, both of which need human involvement to ensure authenticity and accuracy.

- **Invalid Data Scenario:** A user might accidentally enter a number that doesn't match official records. While automated validation could flag this based on format, manual review might be necessary for careless inputs.
- **Control:** Human validation should be integrated into the system workflow. Once the system flags the input as potentially invalid, the data is forwarded for manual review by a qualified agent. This agent can cross-check the information with external verification systems or request additional documentation from the user. This approach ensures that sensitive data is handled with care, minimizing the risk of errors and maintaining user trust in the system's accuracy.

2. Manual Approval for Complex Data

Some data is too complex for automated validation and requires human judgment. Legal documents, contracts, or other high-stakes information often fall into this category, where automated systems may not fully capture the required depth of validation.

- **Example:** Suppose your system processes legal contracts between green suppliers and buyers. Automated systems can verify that all required fields are filled in and that basic formatting is correct, but they cannot interpret the content of the contract to ensure it's valid, enforceable, or free from ambiguity.
- **Invalid Data Scenario:** A user could submit a contract with conflicting clauses, ambiguous terms, or incomplete information. While the system may accept the input based on basic validation, the legal content requires deeper scrutiny to avoid disputes.
- **Control:** Implement a workflow that routes all submitted contracts for human review by legal experts or authorized personnel. These experts can verify that the terms align with sustainability requirements and applicable laws before final approval. This ensures that complex legal data is reviewed thoroughly, preventing issues that automated systems may miss.

3. Preventing Fraud and Ensuring Contextual Awareness

Sensitive information, such as product details information or participant details, is also at risk of fraudulent activity. Automated systems might not detect subtle signs of fraudulent entries, but human reviewers can.

- **Example:** A fraudulent supplier could submit a fake eco-certificate that passes an automated check. However, a human reviewer with knowledge of regional certification standards could detect discrepancies and reject the fraudulent entry.
- **Control:** Incorporating a manual review step for high-risk data fields like certifications and financial data ensures that fraud risks are mitigated. Human reviewers can better understand the context of the data, recognizing unusual patterns or questionable entries that an automated system might miss.

Methods of Human Control

Visual Checking: Involves visually comparing entered data against original documents to detect and correct discrepancies.

Double-Entry: Involves entering data twice, with the system comparing the two entries to identify mismatches. The original source is then checked to determine the correct entry. This method is particularly useful for checking out-of-range values and other subtle errors that may be missed in single-entry systems.

Create and Insert SQL scripts

CREATE

-- MySQL Workbench Forward Engineering

```
SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;
SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;
SET @OLD_SQL_MODE=@@SQL_MODE,
SQL_MODE='ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_D
ATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION';
```

-- Schema mydb

-- Schema greenpackaging

-- Schema greenpackaging

```
CREATE SCHEMA IF NOT EXISTS `greenpackaging` DEFAULT CHARACTER SET utf8mb4 COLLATE
utf8mb4_0900_ai_ci ;
USE `greenpackaging` ;
```

-- Table `greenpackaging`.`sports`

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`sports` (
  `sportID` VARCHAR(3) NOT NULL,
  `sportName` VARCHAR(45) NOT NULL,
  PRIMARY KEY (`sportID`),
  UNIQUE INDEX `sportID_UNIQUE` (`sportID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

-- Table `greenpackaging`.`participant`

```
CREATE TABLE IF NOT EXISTS `greenpackaging`.`participant` (
  `participantID` VARCHAR(10) NOT NULL,
  `participantName` VARCHAR(100) NOT NULL,
  `DOB` DATE NOT NULL,
  `visitDate` DATE NOT NULL,
  `sportID` VARCHAR(3) NOT NULL,
  PRIMARY KEY (`participantID`),
  UNIQUE INDEX `participantID_UNIQUE` (`participantID` ASC) VISIBLE,
  INDEX `participant_sportID_idx` (`sportID` ASC) VISIBLE,
  CONSTRAINT `participant_sportID`
    FOREIGN KEY (`sportID`)
    REFERENCES `greenpackaging`.`sports` (`sportID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```

-----
-- Table `greenpackaging`.`product`
-----
CREATE TABLE IF NOT EXISTS `greenpackaging`.`product` (
  `productID` VARCHAR(4) NOT NULL,
  `qtyProduced` INT NOT NULL,
  `reusabilityStatus` VARCHAR(20) NOT NULL,
  `carbonFootprint` DECIMAL(10,2) NOT NULL,
  CHECK (carbonFootprint >= 0),
  PRIMARY KEY (`productID`),
  UNIQUE INDEX `productID_UNIQUE` (`productID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`supplier`
-----
CREATE TABLE IF NOT EXISTS `greenpackaging`.`supplier` (
  `supplierID` VARCHAR(4) NOT NULL,
  `supplierName` VARCHAR(100) NOT NULL,
  `supplierType` VARCHAR(45) NOT NULL,
  `telNo` VARCHAR(20) NOT NULL,
  PRIMARY KEY (`supplierID`),
  UNIQUE INDEX `supplierID_UNIQUE` (`supplierID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`productsupplier`
-----
CREATE TABLE IF NOT EXISTS `greenpackaging`.`productsupplier` (
  `productID` VARCHAR(4) NOT NULL,
  `supplierID` VARCHAR(4) NOT NULL,
  PRIMARY KEY (`productID`, `supplierID`),
  INDEX `productsupplier_supplierID_idx` (`supplierID` ASC) VISIBLE,
  CONSTRAINT `productsupplier_productID`
    FOREIGN KEY (`productID`)
    REFERENCES `greenpackaging`.`product` (`productID`),
  CONSTRAINT `productsupplier_supplierID`
    FOREIGN KEY (`supplierID`)
    REFERENCES `greenpackaging`.`supplier` (`supplierID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`ratingsystem`
-----
CREATE TABLE IF NOT EXISTS `greenpackaging`.`ratingsystem` (
  `ratingID` VARCHAR(4) NOT NULL,
  `reusabilityStatus` VARCHAR(20) NOT NULL,
  `totalCarbonFootprint` INT NOT NULL,
  `ratingDate` DATE NOT NULL,
  `ratingScore` INT NOT NULL,
  PRIMARY KEY (`ratingID`),

```

```

CHECK (ratingScore >= 0),
UNIQUE INDEX `ratingID_UNIQUE` (`ratingID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`venue`
-----

```

```

CREATE TABLE IF NOT EXISTS `greenpackaging`.`venue` (
  `venueID` VARCHAR(4) NOT NULL,
  `address` VARCHAR(95) NOT NULL,
  `basisInfrastructure` VARCHAR(45) NOT NULL,
  PRIMARY KEY (`venueID`),
  UNIQUE INDEX `venueID_UNIQUE` (`venueID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`ratingprodvenue`
-----

```

```

CREATE TABLE IF NOT EXISTS `greenpackaging`.`ratingprodvenue` (
  `ratingID` VARCHAR(4) NOT NULL,
  `productID` VARCHAR(4) NOT NULL,
  `venueID` VARCHAR(4) NOT NULL,
  PRIMARY KEY (`ratingID`, `productID`, `venueID`),
  INDEX `rpv_productID_idx` (`productID` ASC) VISIBLE,
  INDEX `rpv_venueID_idx` (`venueID` ASC) VISIBLE,
  CONSTRAINT `rpv_productID`
    FOREIGN KEY (`productID`)
      REFERENCES `greenpackaging`.`product` (`productID`),
  CONSTRAINT `rpv_ratingID`
    FOREIGN KEY (`ratingID`)
      REFERENCES `greenpackaging`.`ratingsystem` (`ratingID`),
  CONSTRAINT `rpv_venueID`
    FOREIGN KEY (`venueID`)
      REFERENCES `greenpackaging`.`venue` (`venueID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`service`
-----

```

```

CREATE TABLE IF NOT EXISTS `greenpackaging`.`service` (
  `serviceID` VARCHAR(4) NOT NULL,
  `serviceName` VARCHAR(45) NOT NULL,
  PRIMARY KEY (`serviceID`),
  UNIQUE INDEX `serviceID_UNIQUE` (`serviceID` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`servicevenue`
-----

```

```

-----
CREATE TABLE IF NOT EXISTS `greenpackaging`.`servicevenue` (
  `serviceID` VARCHAR(4) NOT NULL,
  `venueID` VARCHAR(4) NOT NULL,
  PRIMARY KEY (`serviceID`, `venueID`),
  INDEX `servicevenue_venueID_idx` (`venueID` ASC) VISIBLE,
  CONSTRAINT `servicevenue_serviceID`
    FOREIGN KEY (`serviceID`)
      REFERENCES `greenpackaging`.`service` (`serviceID`),
  CONSTRAINT `servicevenue_venueID`
    FOREIGN KEY (`venueID`)
      REFERENCES `greenpackaging`.`venue` (`venueID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`sportproduct`
-----

```

```

CREATE TABLE IF NOT EXISTS `greenpackaging`.`sportproduct` (
  `sportID` VARCHAR(3) NOT NULL,
  `productID` VARCHAR(4) NOT NULL,
  PRIMARY KEY (`sportID`, `productID`),
  INDEX `productID_idx` (`productID` ASC) VISIBLE,
  CONSTRAINT `sportproduct_productID`
    FOREIGN KEY (`productID`)
      REFERENCES `greenpackaging`.`product` (`productID`),
  CONSTRAINT `sportproduct_sportID`
    FOREIGN KEY (`sportID`)
      REFERENCES `greenpackaging`.`sports` (`sportID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

-----
-- Table `greenpackaging`.`venuesport`
-----

```

```

CREATE TABLE IF NOT EXISTS `greenpackaging`.`venuesport` (
  `venueID` VARCHAR(4) NOT NULL,
  `sportID` VARCHAR(3) NOT NULL,
  PRIMARY KEY (`venueID`, `sportID`),
  INDEX `venuesport_sportID_idx` (`sportID` ASC) VISIBLE,
  CONSTRAINT `venuesport_sportID`
    FOREIGN KEY (`sportID`)
      REFERENCES `greenpackaging`.`sports` (`sportID`),
  CONSTRAINT `venuesport_venueID`
    FOREIGN KEY (`venueID`)
      REFERENCES `greenpackaging`.`venue` (`venueID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;

```

```

SET SQL_MODE=@OLD_SQL_MODE;
SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS;
SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS;

```

INSERT

-- First, insert all the “master” tables
-- sports, venue, product, supplier, participant, service

-- MASTER TABLE: sports

```
INSERT INTO sports VALUES ('S01', 'Athletics');
INSERT INTO sports VALUES ('S02', 'Swimming');
INSERT INTO sports VALUES ('S03', 'Gymnastics');
INSERT INTO sports VALUES ('S04', 'Cycling');
INSERT INTO sports VALUES ('S05', 'Basketball');
INSERT INTO sports VALUES ('S06', 'Football');
INSERT INTO sports VALUES ('S07', 'Tennis');
INSERT INTO sports VALUES ('S08', 'Boxing');
INSERT INTO sports VALUES ('S09', 'Weightlifting');
INSERT INTO sports VALUES ('S10', 'Wrestling');
INSERT INTO sports VALUES ('S11', 'Archery');
INSERT INTO sports VALUES ('S12', 'Badminton');
INSERT INTO sports VALUES ('S13', 'Fencing');
INSERT INTO sports VALUES ('S14', 'Diving');
INSERT INTO sports VALUES ('S15', 'Judo');
INSERT INTO sports VALUES ('S16', 'Rowing');
INSERT INTO sports VALUES ('S17', 'Sailing');
INSERT INTO sports VALUES ('S18', 'Volleyball');
INSERT INTO sports VALUES ('S19', 'Rugby');
INSERT INTO sports VALUES ('S20', 'Equestrian');
```

-- MASTER TABLE: venue

```
INSERT INTO venue VALUES ('ST01', 'Brisbane Stadium', 'Land');
INSERT INTO venue VALUES ('ST02', 'Brisbane Aquatics Centre', 'Water');
```

-- MASTER TABLE: product

```
INSERT INTO product VALUES ('P01', 850, 'Non-reusable', 9.5);
INSERT INTO product VALUES ('P02', 589, 'Reusable', 1.5);
INSERT INTO product VALUES ('P03', 436, 'Recyclable', 4);
INSERT INTO product VALUES ('P04', 446, 'Compostable', 3);
INSERT INTO product VALUES ('P05', 690, 'Compostable', 2.3);
INSERT INTO product VALUES ('P06', 784, 'Reusable', 0.3);
INSERT INTO product VALUES ('P07', 917, 'Compostable', 2.7);
INSERT INTO product VALUES ('P08', 221, 'Reusable', 0.1);
INSERT INTO product VALUES ('P09', 680, 'Reusable', 0.2);
INSERT INTO product VALUES ('P10', 632, 'Recyclable', 4.2);
INSERT INTO product VALUES ('P11', 397, 'Non-reusable', 8.8);
INSERT INTO product VALUES ('P12', 487, 'Partially Reusable', 6.3);
INSERT INTO product VALUES ('P13', 165, 'Compostable', 2.2);
INSERT INTO product VALUES ('P14', 311, 'Compostable', 3);
INSERT INTO product VALUES ('P15', 707, 'Compostable', 2.5);
INSERT INTO product VALUES ('P16', 767, 'Recyclable', 5.2);
INSERT INTO product VALUES ('P17', 228, 'Recyclable', 5.7);
INSERT INTO product VALUES ('P18', 489, 'Recyclable', 4.8);
INSERT INTO product VALUES ('P19', 886, 'Compostable', 2);
INSERT INTO product VALUES ('P20', 43, 'Reusable', 0.8);
INSERT INTO product VALUES ('P21', 723, 'Compostable', 2.6);
INSERT INTO product VALUES ('P22', 562, 'Reusable', 1.1);
INSERT INTO product VALUES ('P23', 162, 'Recyclable', 5.9);
```

```

INSERT INTO product VALUES ('P24', 990, 'Reusable', 0.9);
INSERT INTO product VALUES ('P25', 831, 'Reusable', 0.9);
INSERT INTO product VALUES ('P26', 364, 'Reusable', 0.2);
INSERT INTO product VALUES ('P27', 431, 'Partially Reusable', 7.8);
INSERT INTO product VALUES ('P28', 780, 'Recyclable', 5.4);
INSERT INTO product VALUES ('P29', 713, 'Partially Reusable', 7.7);
INSERT INTO product VALUES ('P30', 808, 'Non-reusable', 9.3);

```

-- MASTER TABLE: supplier

```

INSERT INTO supplier VALUES ('SP01', 'Sarina Annable', 'Portion-Control Packaging supplier',
'0432450065');
INSERT INTO supplier VALUES ('SP02', 'Patrizia Kidsley', 'Recycled Packaging supplier', '0412146083');
INSERT INTO supplier VALUES ('SP03', 'Stephi Northrop', 'Food-Grade Packaging supplier', '0491921367');
INSERT INTO supplier VALUES ('SP04', 'Jethro Georgeou', 'Recycled Packaging supplier', '0477271642');
INSERT INTO supplier VALUES ('SP05', 'Edsel Indruch', 'Portion-Control Packaging supplier', '0463911595');
INSERT INTO supplier VALUES ('SP06', 'Dulsea Malby', 'Portion-Control Packaging supplier', '0455320292');
INSERT INTO supplier VALUES ('SP07', 'Nadeen Dome', 'Smart Packaging supplier', '0464829379');
INSERT INTO supplier VALUES ('SP08', 'Geoffrey Deer', 'Recycled Packaging supplier', '0490626112');
INSERT INTO supplier VALUES ('SP09', 'Ferdinand Hicklingbottom', 'Food-Grade Packaging supplier',
'0470506184');
INSERT INTO supplier VALUES ('SP10', 'Christian Gosalvez', 'Food-Grade Packaging supplier',
'0422307640');
INSERT INTO supplier VALUES ('SP11', 'Maegan Robken', 'Smart Packaging supplier', '0428288254');
INSERT INTO supplier VALUES ('SP12', 'Gilbertina Padefield', 'Food-Grade Packaging supplier',
'0476015094');
INSERT INTO supplier VALUES ('SP13', 'Giraldo Fellows', 'Smart Packaging supplier', '0443516570');
INSERT INTO supplier VALUES ('SP14', 'Parsifal Clayworth', 'Smart Packaging supplier', '0492183573');
INSERT INTO supplier VALUES ('SP15', 'Sherlock Wringe', 'Recycled Packaging supplier', '0458457118');
INSERT INTO supplier VALUES ('SP16', 'Ronnie Lafontaine', 'Smart Packaging supplier', '0482223325');
INSERT INTO supplier VALUES ('SP17', 'Irene Treend', 'Food-Grade Packaging supplier', '0470169927');
INSERT INTO supplier VALUES ('SP18', 'Rickie Sleit', 'Smart Packaging supplier', '0453107666');
INSERT INTO supplier VALUES ('SP19', 'Noel Skillern', 'Recycled Packaging supplier', '0450253499');
INSERT INTO supplier VALUES ('SP20', 'Travis Thumnel', 'Portion-Control Packaging supplier',
'0453162861');

```

-- MASTER TABLE: participant

```

INSERT INTO participant VALUES ('P01', 'Isac Culshaw', '2004-08-26', '2032-07-29', 'S04');
INSERT INTO participant VALUES ('P02', 'Tatiana Scarce', '1999-11-17', '2032-08-05', 'S16');
INSERT INTO participant VALUES ('P03', 'Alanna Fenney', '1997-12-28', '2032-07-25', 'S06');
INSERT INTO participant VALUES ('P04', 'Eugenie VanBrugh', '1998-03-31', '2032-07-26', 'S11');
INSERT INTO participant VALUES ('P05', 'Curt Hamberstone', '2003-11-14', '2032-08-02', 'S13');
INSERT INTO participant VALUES ('P06', 'Simeon Karpinski', '1980-11-24', '2032-08-05', 'S07');
INSERT INTO participant VALUES ('P07', 'Maressa Jerrand', '2004-06-09', '2032-08-03', 'S11');
INSERT INTO participant VALUES ('P08', 'Brittan Major', '1999-11-15', '2032-07-24', 'S12');
INSERT INTO participant VALUES ('P09', 'Joe Heathorn', '1982-02-14', '2032-08-03', 'S02');
INSERT INTO participant VALUES ('P10', 'Katuscha Fencott', '1993-10-09', '2032-07-31', 'S10');
INSERT INTO participant VALUES ('P11', 'Brittney Evitts', '1984-02-10', '2032-07-23', 'S09');
INSERT INTO participant VALUES ('P12', 'Sigfried Verzey', '2005-05-03', '2032-08-07', 'S19');
INSERT INTO participant VALUES ('P13', 'Jodie Chisman', '1990-05-05', '2032-07-24', 'S05');
INSERT INTO participant VALUES ('P14', 'Denney Breen', '2006-04-07', '2032-07-31', 'S16');
INSERT INTO participant VALUES ('P15', 'Husain Maryman', '2006-06-14', '2032-08-05', 'S03');
INSERT INTO participant VALUES ('P16', 'Danika Woodnutt', '2006-05-06', '2032-08-07', 'S01');
INSERT INTO participant VALUES ('P17', 'Cyrill Woofinden', '1988-06-05', '2032-07-23', 'S07');
INSERT INTO participant VALUES ('P18', 'Christopher Hardisty', '1981-04-13', '2032-08-04', 'S14');
INSERT INTO participant VALUES ('P19', 'Carleen Loudwell', '1984-03-07', '2032-07-30', 'S08');
INSERT INTO participant VALUES ('P20', 'Wilbur Gabriely', '2004-07-11', '2032-07-25', 'S20');
INSERT INTO participant VALUES ('P21', 'Mendie Bick', '1997-10-29', '2032-08-02', 'S15');
INSERT INTO participant VALUES ('P22', 'Allys Lidgertwood', '1988-12-10', '2032-07-23', 'S09');

```

```

INSERT INTO participant VALUES ('P23', 'Teodoro Leuty', '2005-01-04', '2032-07-28', 'S13');
INSERT INTO participant VALUES ('P24', 'Hersch Diggles', '2003-07-13', '2032-08-06', 'S17');
INSERT INTO participant VALUES ('P25', 'Thomasina Hartness', '1982-05-09', '2032-07-31', 'S08');
INSERT INTO participant VALUES ('P26', 'John Blesing', '1984-10-18', '2032-07-31', 'S01');
INSERT INTO participant VALUES ('P27', 'Allen Zorzoni', '2005-02-05', '2032-07-26', 'S18');
INSERT INTO participant VALUES ('P28', 'Bar Solly', '1990-05-19', '2032-07-28', 'S12');
INSERT INTO participant VALUES ('P29', 'Hobie Grieves', '1982-01-20', '2032-07-26', 'S16');
INSERT INTO participant VALUES ('P30', 'Zorina Jindacek', '1986-07-05', '2032-07-30', 'S04');
INSERT INTO participant VALUES ('P31', 'Constancy Kohler', '1992-10-30', '2032-08-05', 'S05');
INSERT INTO participant VALUES ('P32', 'Pete Measey', '1984-08-05', '2032-07-26', 'S20');
INSERT INTO participant VALUES ('P33', 'Katya Cawsby', '1996-12-05', '2032-07-27', 'S02');
INSERT INTO participant VALUES ('P34', 'Roseann Ickov', '1998-11-11', '2032-08-04', 'S15');
INSERT INTO participant VALUES ('P35', 'Malynda Blagden', '1993-07-16', '2032-07-30', 'S12');
INSERT INTO participant VALUES ('P36', 'Claudio Kall', '1989-12-20', '2032-07-30', 'S03');
INSERT INTO participant VALUES ('P37', 'Wernher Bremen', '2004-03-04', '2032-08-03', 'S07');
INSERT INTO participant VALUES ('P38', 'Mario Crofthwaite', '1982-09-06', '2032-08-02', 'S10');
INSERT INTO participant VALUES ('P39', 'Xenia Lohoar', '2004-09-24', '2032-07-26', 'S05');
INSERT INTO participant VALUES ('P40', 'Kacy Hawkshaw', '2003-10-26', '2032-07-28', 'S19');
INSERT INTO participant VALUES ('P41', 'Waldon Oleszcuk', '1989-04-20', '2032-07-29', 'S08');
INSERT INTO participant VALUES ('P42', 'Vaughan Arnald', '1995-03-12', '2032-07-23', 'S18');
INSERT INTO participant VALUES ('P43', 'Eolanda Morphey', '2002-09-09', '2032-07-31', 'S11');
INSERT INTO participant VALUES ('P44', 'Julee Hubbert', '2005-07-20', '2032-07-24', 'S10');
INSERT INTO participant VALUES ('P45', 'Cissiee Geipel', '1985-01-10', '2032-07-31', 'S09');
INSERT INTO participant VALUES ('P46', 'Edwin Guerrieri', '1987-04-10', '2032-07-31', 'S14');
INSERT INTO participant VALUES ('P47', 'Dori Pennington', '1982-10-31', '2032-07-30', 'S17');
INSERT INTO participant VALUES ('P48', 'Gualterio Middiff', '1991-01-31', '2032-08-03', 'S13');
INSERT INTO participant VALUES ('P49', 'Pavla Surpliss', '1994-01-05', '2032-07-27', 'S06');
INSERT INTO participant VALUES ('P50', 'Kiley Cawsy', '2004-03-17', '2032-07-23', 'S06');

```

```
-- MASTER TABLE: service
```

```

INSERT INTO service VALUES ('SE1', 'Waste collection');
INSERT INTO service VALUES ('SE2', 'Sanitisation');
INSERT INTO service VALUES ('SE3', 'Hospitality');
INSERT INTO service VALUES ('SE4', 'Medical & health');
INSERT INTO service VALUES ('SE5', 'Field & equipment cleaning & maintenance');

```

```

-- -----
-- Then, insert all the "child" tables
-- sportproduct, venuesport, productsupplier,
-- servicevenue, ratingsystem, ratingprodvenue
-- -----

```

```
-- TABLE: sportproduct
```

```

INSERT INTO sportproduct VALUES ('S01', 'P08');
INSERT INTO sportproduct VALUES ('S01', 'P02');
INSERT INTO sportproduct VALUES ('S01', 'P03');
INSERT INTO sportproduct VALUES ('S01', 'P22');
INSERT INTO sportproduct VALUES ('S01', 'P07');
INSERT INTO sportproduct VALUES ('S02', 'P04');
INSERT INTO sportproduct VALUES ('S02', 'P09');
INSERT INTO sportproduct VALUES ('S02', 'P21');
INSERT INTO sportproduct VALUES ('S02', 'P08');
INSERT INTO sportproduct VALUES ('S02', 'P30');
INSERT INTO sportproduct VALUES ('S03', 'P10');
INSERT INTO sportproduct VALUES ('S03', 'P17');
INSERT INTO sportproduct VALUES ('S03', 'P22');
INSERT INTO sportproduct VALUES ('S03', 'P01');

```

```

INSERT INTO sportproduct VALUES ('S03', 'P14');
INSERT INTO sportproduct VALUES ('S04', 'P06');
INSERT INTO sportproduct VALUES ('S04', 'P25');
INSERT INTO sportproduct VALUES ('S04', 'P12');
INSERT INTO sportproduct VALUES ('S04', 'P16');
INSERT INTO sportproduct VALUES ('S04', 'P13');
INSERT INTO sportproduct VALUES ('S05', 'P21');
INSERT INTO sportproduct VALUES ('S05', 'P02');
INSERT INTO sportproduct VALUES ('S05', 'P26');
INSERT INTO sportproduct VALUES ('S05', 'P13');
INSERT INTO sportproduct VALUES ('S05', 'P17');
INSERT INTO sportproduct VALUES ('S06', 'P19');
INSERT INTO sportproduct VALUES ('S06', 'P07');
INSERT INTO sportproduct VALUES ('S06', 'P29');
INSERT INTO sportproduct VALUES ('S06', 'P12');
INSERT INTO sportproduct VALUES ('S06', 'P26');
INSERT INTO sportproduct VALUES ('S07', 'P30');
INSERT INTO sportproduct VALUES ('S07', 'P08');
INSERT INTO sportproduct VALUES ('S07', 'P03');
INSERT INTO sportproduct VALUES ('S07', 'P06');
INSERT INTO sportproduct VALUES ('S07', 'P10');
INSERT INTO sportproduct VALUES ('S08', 'P04');
INSERT INTO sportproduct VALUES ('S08', 'P15');
INSERT INTO sportproduct VALUES ('S08', 'P07');
INSERT INTO sportproduct VALUES ('S08', 'P11');
INSERT INTO sportproduct VALUES ('S08', 'P19');
INSERT INTO sportproduct VALUES ('S09', 'P17');
INSERT INTO sportproduct VALUES ('S09', 'P27');
INSERT INTO sportproduct VALUES ('S09', 'P06');
INSERT INTO sportproduct VALUES ('S09', 'P29');
INSERT INTO sportproduct VALUES ('S09', 'P22');
INSERT INTO sportproduct VALUES ('S10', 'P25');
INSERT INTO sportproduct VALUES ('S10', 'P08');
INSERT INTO sportproduct VALUES ('S10', 'P18');
INSERT INTO sportproduct VALUES ('S10', 'P13');
INSERT INTO sportproduct VALUES ('S10', 'P30');
INSERT INTO sportproduct VALUES ('S11', 'P24');
INSERT INTO sportproduct VALUES ('S11', 'P03');
INSERT INTO sportproduct VALUES ('S11', 'P27');
INSERT INTO sportproduct VALUES ('S11', 'P25');
INSERT INTO sportproduct VALUES ('S11', 'P23');
INSERT INTO sportproduct VALUES ('S12', 'P21');
INSERT INTO sportproduct VALUES ('S12', 'P16');
INSERT INTO sportproduct VALUES ('S12', 'P01');
INSERT INTO sportproduct VALUES ('S12', 'P07');
INSERT INTO sportproduct VALUES ('S12', 'P24');
INSERT INTO sportproduct VALUES ('S13', 'P09');
INSERT INTO sportproduct VALUES ('S13', 'P02');
INSERT INTO sportproduct VALUES ('S13', 'P20');
INSERT INTO sportproduct VALUES ('S13', 'P17');
INSERT INTO sportproduct VALUES ('S13', 'P18');
INSERT INTO sportproduct VALUES ('S14', 'P29');
INSERT INTO sportproduct VALUES ('S14', 'P10');
INSERT INTO sportproduct VALUES ('S14', 'P28');
INSERT INTO sportproduct VALUES ('S14', 'P20');
INSERT INTO sportproduct VALUES ('S14', 'P16');
INSERT INTO sportproduct VALUES ('S15', 'P20');
INSERT INTO sportproduct VALUES ('S15', 'P11');
INSERT INTO sportproduct VALUES ('S15', 'P28');
INSERT INTO sportproduct VALUES ('S15', 'P18');

```

```

INSERT INTO sportproduct VALUES ('S15', 'P23');
INSERT INTO sportproduct VALUES ('S16', 'P17');
INSERT INTO sportproduct VALUES ('S16', 'P27');
INSERT INTO sportproduct VALUES ('S16', 'P11');
INSERT INTO sportproduct VALUES ('S16', 'P14');
INSERT INTO sportproduct VALUES ('S16', 'P04');
INSERT INTO sportproduct VALUES ('S17', 'P09');
INSERT INTO sportproduct VALUES ('S17', 'P28');
INSERT INTO sportproduct VALUES ('S17', 'P23');
INSERT INTO sportproduct VALUES ('S17', 'P17');
INSERT INTO sportproduct VALUES ('S17', 'P11');
INSERT INTO sportproduct VALUES ('S18', 'P05');
INSERT INTO sportproduct VALUES ('S18', 'P01');
INSERT INTO sportproduct VALUES ('S18', 'P06');
INSERT INTO sportproduct VALUES ('S18', 'P09');
INSERT INTO sportproduct VALUES ('S18', 'P15');
INSERT INTO sportproduct VALUES ('S19', 'P07');
INSERT INTO sportproduct VALUES ('S19', 'P02');
INSERT INTO sportproduct VALUES ('S19', 'P05');
INSERT INTO sportproduct VALUES ('S19', 'P24');
INSERT INTO sportproduct VALUES ('S19', 'P09');
INSERT INTO sportproduct VALUES ('S20', 'P04');
INSERT INTO sportproduct VALUES ('S20', 'P01');
INSERT INTO sportproduct VALUES ('S20', 'P10');
INSERT INTO sportproduct VALUES ('S20', 'P05');
INSERT INTO sportproduct VALUES ('S20', 'P26');

```

-- TABLE: venuesport

```

INSERT INTO venuesport VALUES ('ST01', 'S01');
INSERT INTO venuesport VALUES ('ST01', 'S03');
INSERT INTO venuesport VALUES ('ST01', 'S04');
INSERT INTO venuesport VALUES ('ST01', 'S05');
INSERT INTO venuesport VALUES ('ST01', 'S06');
INSERT INTO venuesport VALUES ('ST01', 'S07');
INSERT INTO venuesport VALUES ('ST01', 'S08');
INSERT INTO venuesport VALUES ('ST01', 'S09');
INSERT INTO venuesport VALUES ('ST01', 'S10');
INSERT INTO venuesport VALUES ('ST01', 'S11');
INSERT INTO venuesport VALUES ('ST01', 'S12');
INSERT INTO venuesport VALUES ('ST01', 'S13');
INSERT INTO venuesport VALUES ('ST01', 'S15');
INSERT INTO venuesport VALUES ('ST01', 'S18');
INSERT INTO venuesport VALUES ('ST01', 'S19');
INSERT INTO venuesport VALUES ('ST01', 'S20');
INSERT INTO venuesport VALUES ('ST02', 'S02');
INSERT INTO venuesport VALUES ('ST02', 'S14');
INSERT INTO venuesport VALUES ('ST02', 'S16');
INSERT INTO venuesport VALUES ('ST02', 'S17');

```

-- TABLE: productsupplier

```

INSERT INTO productsupplier VALUES ('P01', 'SP14');
INSERT INTO productsupplier VALUES ('P02', 'SP03');
INSERT INTO productsupplier VALUES ('P03', 'SP02');
INSERT INTO productsupplier VALUES ('P04', 'SP10');
INSERT INTO productsupplier VALUES ('P05', 'SP04');
INSERT INTO productsupplier VALUES ('P06', 'SP12');
INSERT INTO productsupplier VALUES ('P07', 'SP17');
INSERT INTO productsupplier VALUES ('P08', 'SP06');

```

```

INSERT INTO productsupplier VALUES ('P09', 'SP08');
INSERT INTO productsupplier VALUES ('P10', 'SP10');
INSERT INTO productsupplier VALUES ('P11', 'SP16');
INSERT INTO productsupplier VALUES ('P12', 'SP01');
INSERT INTO productsupplier VALUES ('P13', 'SP09');
INSERT INTO productsupplier VALUES ('P14', 'SP06');
INSERT INTO productsupplier VALUES ('P15', 'SP18');
INSERT INTO productsupplier VALUES ('P16', 'SP04');
INSERT INTO productsupplier VALUES ('P17', 'SP19');
INSERT INTO productsupplier VALUES ('P18', 'SP11');
INSERT INTO productsupplier VALUES ('P19', 'SP09');
INSERT INTO productsupplier VALUES ('P20', 'SP13');
INSERT INTO productsupplier VALUES ('P21', 'SP07');
INSERT INTO productsupplier VALUES ('P22', 'SP07');
INSERT INTO productsupplier VALUES ('P23', 'SP02');
INSERT INTO productsupplier VALUES ('P24', 'SP05');
INSERT INTO productsupplier VALUES ('P25', 'SP01');
INSERT INTO productsupplier VALUES ('P26', 'SP20');
INSERT INTO productsupplier VALUES ('P27', 'SP15');
INSERT INTO productsupplier VALUES ('P28', 'SP03');
INSERT INTO productsupplier VALUES ('P29', 'SP08');
INSERT INTO productsupplier VALUES ('P30', 'SP05');

```

-- TABLE: servicevenue

```

INSERT INTO servicevenue VALUES ('SE1', 'ST01');
INSERT INTO servicevenue VALUES ('SE2', 'ST01');
INSERT INTO servicevenue VALUES ('SE3', 'ST01');
INSERT INTO servicevenue VALUES ('SE4', 'ST01');
INSERT INTO servicevenue VALUES ('SE5', 'ST01');
INSERT INTO servicevenue VALUES ('SE1', 'ST02');
INSERT INTO servicevenue VALUES ('SE2', 'ST02');
INSERT INTO servicevenue VALUES ('SE3', 'ST02');
INSERT INTO servicevenue VALUES ('SE4', 'ST02');
INSERT INTO servicevenue VALUES ('SE5', 'ST02');

```

-- TABLE: ratingsystem

```

INSERT INTO ratingsystem VALUES ('R01', 'Reusable', 1.5, '2032-08-03', 14);
INSERT INTO ratingsystem VALUES ('R02', 'Recyclable', 4, '2032-08-01', 9);
INSERT INTO ratingsystem VALUES ('R03', 'Compostable', 2.3, '2032-08-01', 10);
INSERT INTO ratingsystem VALUES ('R04', 'Reusable', 0.3, '2032-08-05', 15);
INSERT INTO ratingsystem VALUES ('R05', 'Compostable', 2.7, '2032-08-06', 10);
INSERT INTO ratingsystem VALUES ('R06', 'Reusable', 0.2, '2032-08-07', 13);
INSERT INTO ratingsystem VALUES ('R07', 'Recyclable', 4.2, '2032-08-05', 9);
INSERT INTO ratingsystem VALUES ('R08', 'Non-reusable', 8.8, '2032-08-06', 1);
INSERT INTO ratingsystem VALUES ('R09', 'Compostable', 3, '2032-08-06', 11);
INSERT INTO ratingsystem VALUES ('R10', 'Compostable', 2.5, '2032-08-02', 10);
INSERT INTO ratingsystem VALUES ('R11', 'Recyclable', 5.2, '2032-08-01', 8);
INSERT INTO ratingsystem VALUES ('R12', 'Recyclable', 5.7, '2032-08-04', 9);
INSERT INTO ratingsystem VALUES ('R13', 'Recyclable', 4.8, '2032-08-06', 7);
INSERT INTO ratingsystem VALUES ('R14', 'Reusable', 0.8, '2032-08-03', 14);
INSERT INTO ratingsystem VALUES ('R15', 'Compostable', 2.6, '2032-08-06', 10);
INSERT INTO ratingsystem VALUES ('R16', 'Reusable', 1.1, '2032-08-03', 14);
INSERT INTO ratingsystem VALUES ('R17', 'Recyclable', 5.9, '2032-08-03', 7);
INSERT INTO ratingsystem VALUES ('R18', 'Reusable', 0.9, '2032-08-01', 14);
INSERT INTO ratingsystem VALUES ('R19', 'Reusable', 0.9, '2032-08-05', 13);
INSERT INTO ratingsystem VALUES ('R20', 'Partially Reusable', 7.8, '2032-08-07', 5);
INSERT INTO ratingsystem VALUES ('R21', 'Recyclable', 5.4, '2032-08-03', 9);

```

```

INSERT INTO ratingsystem VALUES ('R22', 'Partially Reusable', 7.7, '2032-08-02', 6);
INSERT INTO ratingsystem VALUES ('R23', 'Non-reusable', 9.5, '2032-08-07', 2);
INSERT INTO ratingsystem VALUES ('R24', 'Reusable', 1.5, '2032-08-02', 15);
INSERT INTO ratingsystem VALUES ('R25', 'Recyclable', 4, '2032-08-06', 8);
INSERT INTO ratingsystem VALUES ('R26', 'Compostable', 3, '2032-08-01', 10);
INSERT INTO ratingsystem VALUES ('R27', 'Compostable', 2.7, '2032-08-03', 12);
INSERT INTO ratingsystem VALUES ('R28', 'Reusable', 0.1, '2032-08-07', 13);
INSERT INTO ratingsystem VALUES ('R29', 'Recyclable', 4.2, '2032-08-01', 7);
INSERT INTO ratingsystem VALUES ('R30', 'Non-reusable', 8.8, '2032-08-04', 1);
INSERT INTO ratingsystem VALUES ('R31', 'Partially Reusable', 6.3, '2032-08-03', 4);
INSERT INTO ratingsystem VALUES ('R32', 'Compostable', 2.2, '2032-08-05', 11);
INSERT INTO ratingsystem VALUES ('R33', 'Compostable', 3, '2032-08-05', 10);
INSERT INTO ratingsystem VALUES ('R34', 'Recyclable', 5.7, '2032-08-05', 7);
INSERT INTO ratingsystem VALUES ('R35', 'Recyclable', 4.8, '2032-08-05', 7);
INSERT INTO ratingsystem VALUES ('R36', 'Compostable', 2, '2032-08-05', 11);
INSERT INTO ratingsystem VALUES ('R37', 'Reusable', 0.8, '2032-08-01', 15);
INSERT INTO ratingsystem VALUES ('R38', 'Recyclable', 5.9, '2032-08-05', 9);
INSERT INTO ratingsystem VALUES ('R39', 'Reusable', 0.9, '2032-08-07', 13);
INSERT INTO ratingsystem VALUES ('R40', 'Reusable', 0.9, '2032-08-03', 14);
INSERT INTO ratingsystem VALUES ('R41', 'Reusable', 0.2, '2032-08-04', 13);
INSERT INTO ratingsystem VALUES ('R42', 'Partially Reusable', 7.8, '2032-08-06', 5);
INSERT INTO ratingsystem VALUES ('R43', 'Partially Reusable', 7.7, '2032-08-07', 5);
INSERT INTO ratingsystem VALUES ('R44', 'Non-reusable', 9.3, '2032-08-05', 2);

```

-- TABLE: ratingprodvenue

```

INSERT INTO ratingprodvenue VALUES ('R01', 'P02', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R02', 'P03', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R03', 'P05', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R04', 'P06', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R05', 'P07', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R06', 'P09', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R07', 'P10', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R08', 'P11', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R09', 'P14', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R10', 'P15', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R11', 'P16', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R12', 'P17', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R13', 'P18', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R14', 'P20', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R15', 'P21', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R16', 'P22', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R17', 'P23', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R18', 'P24', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R19', 'P25', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R20', 'P27', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R21', 'P28', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R22', 'P29', 'ST01');
INSERT INTO ratingprodvenue VALUES ('R23', 'P01', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R24', 'P02', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R25', 'P03', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R26', 'P04', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R27', 'P07', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R28', 'P08', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R29', 'P10', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R30', 'P11', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R31', 'P12', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R32', 'P13', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R33', 'P14', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R34', 'P17', 'ST02');

```

```
INSERT INTO ratingprodvenue VALUES ('R35', 'P18', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R36', 'P19', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R37', 'P20', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R38', 'P23', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R39', 'P24', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R40', 'P25', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R41', 'P26', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R42', 'P27', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R43', 'P29', 'ST02');
INSERT INTO ratingprodvenue VALUES ('R44', 'P30', 'ST02');
```

Explanation of Security Controls

Security Control

Database security encompasses the processes, tools, and controls designed to safeguard databases against accidental or intentional threats. The primary objectives are to protect sensitive data and ensure the confidentiality, availability, and integrity of the database (Microsoft, 2024). For our green purchasing DBMS, the data includes crucial sustainability metrics such as product comparisons and the carbon footprint of products and services. Unauthorized access to this data could not only lead to its misuse but also compromise the integrity of sustainability initiatives and cause reputational harm to stakeholders.

Database security helps to secure database management systems from malicious cyber – attacks and illegitimate use while it also not only protects the data but also avoids misuse damage and intrusion (Imperva, 2021). To protect our Database management system, we plan to go through a combination of physical access controls, network security, and DBMS security. These controls not only protect the data but ensure that only authorized users can access the system, maintaining both confidentiality and accuracy, thus supporting our goal of facilitating sustainable and ethical decision-making.

1. Physical Access Controls (Physical Database security)

Definition and Purpose

Physical access controls refer to the security measures that restrict physical access to the database servers and storage devices. These controls are crucial in preventing unauthorized individuals from physically accessing the hardware that hosts the database, thereby protecting it from theft, damage, or unauthorized modification.

Techniques and Implementation

- **Secure Server Rooms:** Implementing secure server rooms with controlled access is fundamental. This involves using locks, keycard access systems, and biometric authentication to ensure that only authorized personnel can enter. For instance, in our green purchasing DBMS, access to the server room is restricted to IT personnel and database administrators only, minimizing the risk of physical breaches.
- **CCTV Monitoring:** 24/7 CCTV monitoring can be used to secure the server room area of the data center. The key of entering the room should be held by the authorised person and all the cameras should be set at all entrance points. (*Physical & Data Security Controls | Enable Help Center, 2024*)
- **Environmental Security:** Maintaining an appropriate environment (temperature, humidity) within server rooms by installing fire, heat and smoke detection is essential to protect hardware. These controls prevent hardware failures and destruction of data centres. (*Physical & Data Security Controls | Enable Help Center, 2024*)

2. Network Security

Definition and Purpose

Network security encompasses the policies, practices, and tools designed to protect the integrity, confidentiality, and accessibility of computer networks and data. Effective network

security prevents unauthorized access to the network that hosts the database and protects data from interception during transmission.

Techniques and Implementation

- **Firewalls:** Firewalls serve as a barrier between the internal network and external threats. They monitor incoming and outgoing traffic based on predetermined security rules. In our green purchasing DBMS, a firewall is configured to block unauthorized traffic while allowing legitimate data requests from verified users.
- **Intrusion Detection and Prevention Systems (IDPS):** IDPS monitors network traffic for suspicious activity and can take action, such as blocking potential threats in real time. For instance, if an unauthorized IP address attempts to access the database, the IDPS will log this attempt and alert the security team.

3. DBMS Security

Definition and Purpose

DBMS security involves the measures and protocols implemented within the database management system itself to protect the data it contains. This includes user authentication, authorization, auditing mechanisms, and data recovery strategies designed to prevent unauthorized access while ensuring data availability for legitimate users.

Techniques and Implementation

- **User Authentication and Authorization:** Establishing strict user authentication processes via physical device, such as fingerprint recognition, face recognition and retina scan. In our green purchasing DBMS, all users must undergo these authentication processes to gain access to sensitive data, such as carbon footprint reports and supplier compliance metrics.
- **Access Control Policies:** Defining user roles and permissions helps limit the access to data based on the user's position in the organization. For instance, only administrative users can modify product data, while other users may only have read access to certain metrics.
- **Data Encryption:** In our DBMS, all sustainability data, including reports on environmental impact, is encrypted to ensure that even if data is intercepted, it cannot be accessed without the proper decryption keys.
- **Audit Trails:** Maintaining audit logs of all database transactions allows us to monitor access and changes made to the database. This is crucial for identifying and responding to suspicious activities, ensuring accountability, and maintaining data integrity.
- **Passwords:** Passwords are an important element in securing DBMS security. It provides authorised users with a short and unique password with the permissions they need every time when they need to gain access to the database (Imperva, 2021). The other function keeps monitoring the activities completed during the working period and stops administrators from sharing their passwords. Imperva (2021) also stated that all passwords are not permitted to be shared, and the account should avoid multiple login attempts and lock it when someone violates the rules.
- **Data Recovery and Backup:** Effective backup and recovery strategies helps to protect organizations data, and it allows recovery quickly when there is a system failure, cyberattack or accidental data loss occurred. For example,

1. Securing databases in the cloud is a method that reduces cost, allow staff for more important work. Although there is some risk by using this method such as extended network perimeter, it can provide better security than most organizations have on-premises (*Database Security Best Practices*, 2020).
2. Journalizing: It is the process of keeping and maintaining a log file of all changes made to the database to enable recovery to be undertaken effectively in the event of a failure.
3. Full backup is a typical backup procedure that involves full backups of the entire database in a specific period such as weekly basis or backup per three days.

Data Cleaning and Data Quality Maintenance Plan

Identifying major data quality problems

(1) Quantitative data problems: confusions, different ranges, non-significant values, outliers, duplicate records, different numerical scales, incorrect data types

(2) Qualitative data problems: synonyms, homonyms, misspelt data, abbreviations, multiple representation, domain violations, irrelevant data, inconsistency, lexical errors, wrong data format (Chen et al., 2019)

(3) All data types' problems: missing values, domain violations, redundancy, inconsistency, data dimension problem (Georgieva et al., 2020)

Data cleaning procedures

Repeated cycles of screening:

This step aims to identify the types of errors and inconsistencies that need to be removed, so a detailed screening is essential. In addition to a manual inspection of the above data quality problems, data analysis tools should be used to gain metadata about the data properties and detect the errors left after manual inspection. (Georgieva et al., 2020)

There are two main approaches for data analysis: data profiling and data mining. Data profiling focuses on the instance analysis of individual attributes. It derives information such as the data type, length, uniqueness, null values, etc., providing an exact view of various quality aspects of the attribute (Rahm & Do, 2000). For example, ensuring attributes like participantID, noParticipant has the correct type of data (e.g., noParticipant should be an integer), ensuring key attributes like sportID and productID are unique, preventing duplicate records. And ensuring venueID in the Venue table is never null. Data mining helps discover specific data patterns in large data sets, e.g., relationships holding between several attributes (Rahm & Do, 2000). This could involve exploring relationships such as Product and Sponsor where examining the relationships between productID, sponsorID, and reusabilityStatus to detect which sponsors contribute to more sustainable practices by providing reusable products.

Conflict resolution:

Attribute split: Free-form attributes often contain multiple values that should be extracted to achieve a more accurate representation and prepare for the further cleaning steps like instance matching and duplicate elimination. For example, the address field in the Venue table, which could contain multiple components (e.g. street, city, suburb) that should be separate or reordered within a field to resolve issues like word transpositions for better data management. (Rahm & Do, 2000)

Validation and correction: We want this step to be conducted automatically as much as possible, involving reviewing each data instance to identify and correct any data entry errors. We may make use of digital dictionary lookups to check the spelling and fix possible mistakes. For example, dictionaries on geographic names and zip codes help to correct address data. Attribute dependencies can be utilized to detect problems and substitute missing values or correct wrong values. (Rahm & Do, 2000)

Standardization: To facilitate instance matching and integration, attribute values should be converted to a consistent and uniform format (Rahm & Do, 2000). For example, dates in fields like DOB in Participant table should follow a consistent format (e.g., YYYY-MM-DD), and text fields such as participantName or venueName should be uniformly converted to either all uppercase or all lowercase to maintain consistency across the database. Additionally, text data can be simplified by applying techniques like stemming, as well as

removing prefixes, suffixes, and common stop words to ensure uniformity across the dataset. (Rahm & Do, 2000)

Verification:

The correctness and effectiveness of a transformation workflow and the transformation definitions should be tested and evaluated, e.g., on a sample or copy of the source data, to improve the definitions if necessary. Multiple iterations of the analysis, design and verification steps may be needed (Karrar & Ali, 2016). For example, some certain data issues in our system like incorrect carbonFootprint values in the Product table, may only become noticeable after specific transformations or calculations, such as aggregating data across different venues or sponsors.

Backflow of cleaned data:

Once all problems are resolved, the cleaned data should also cover the dirty data in the original sources in order to give legacy applications the improved data too and to avoid redoing the cleaning work for future data extractions. For data warehousing, the cleaned data is available from the data staging area. (Karrar & Ali, 2016)

Data cleaning plan		
Process	Responsible person	Time arrangement
Screening-manual inspection	Data Analyst 1	Daily or every two days based on the importance of the data
Screening-use analysis programs	Data Analyst 2	Automated analysis programs can be set to run every hour
Conflict resolution-Attribute split	Data Engineer 1	Every two hours during real-time data input
Conflict resolution-Validation and correction	Data Engineer 2	Immediately after data input/every hour
Conflict resolution-Standardization	Data Engineer 3	Every hour
Verification	Data Analyst 1	Every two hours (depending on the volume of data)
Backflow of cleaned data	Data Engineer 1	Daily

Useful reports

Report 1: Product Options Report

Code:

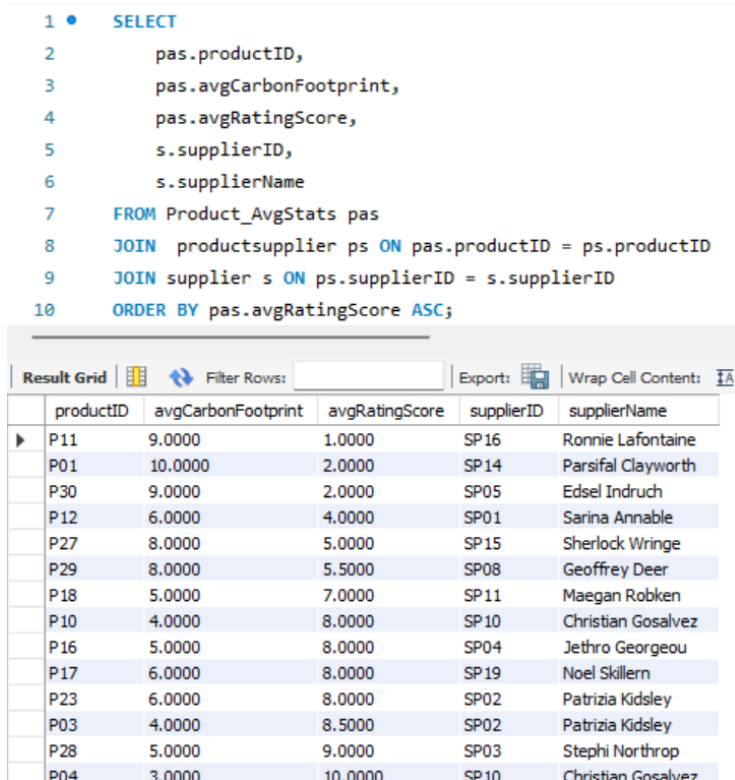
Creating the view:

```
CREATE VIEW Product_AvgStats AS
SELECT
    rpv.productID,
    AVG(rate.totalCarbonFootprint) AS avgCarbonFootprint,
    AVG(rate.ratingScore) AS avgRatingScore
FROM ratingprodvenue rpv
JOIN ratingsystem rate ON rpv.ratingID = rate.ratingID
GROUP BY rpv.productID;
```

Using the view created to run the query:

```
SELECT
    pas.productID,
    pas.avgCarbonFootprint,
    pas.avgRatingScore,
    s.supplierID,
    s.supplierName
FROM Product_AvgStats pas
JOIN productsupplier ps ON pas.productID = ps.productID
JOIN supplier s ON ps.supplierID = s.supplierID
ORDER BY pas.avgRatingScore ASC;
```

Screenshot:



The screenshot shows a SQL query editor with a query that selects product details and average statistics from a view. Below the query is a 'Result Grid' showing the output of the query. The grid has columns for productID, avgCarbonFootprint, avgRatingScore, supplierID, and supplierName. The results are sorted by avgRatingScore in ascending order.

productID	avgCarbonFootprint	avgRatingScore	supplierID	supplierName
P11	9.0000	1.0000	SP16	Ronnie Lafontaine
P01	10.0000	2.0000	SP14	Parsifal Clayworth
P30	9.0000	2.0000	SP05	Edsel Indruch
P12	6.0000	4.0000	SP01	Sarina Annable
P27	8.0000	5.0000	SP15	Sherlock Wringe
P29	8.0000	5.5000	SP08	Geoffrey Deer
P18	5.0000	7.0000	SP11	Maegan Robken
P10	4.0000	8.0000	SP10	Christian Gosalvez
P16	5.0000	8.0000	SP04	Jethro Georgeou
P17	6.0000	8.0000	SP19	Noel Skillern
P23	6.0000	8.0000	SP02	Patrizia Kidsley
P03	4.0000	8.5000	SP02	Patrizia Kidsley
P28	5.0000	9.0000	SP03	Stephi Northrop
P04	3.0000	10.0000	SP10	Christian Gosalvez

This SQL query calculates the average carbon footprint and average rating score (with higher carbon footprint indicating more harm) for each product associated with various ratings provided at different venues.

This query is meaningful as it provides **sustainability insight** by identifying which products are more environmentally friendly. Products with a lower average carbon footprint align better with sustainability goals, making them more desirable for eco-conscious procurement. This query provides **quality assessment** against the quality of each product based on user feedback to determine which products are most favored or effective in practice. And the query also provides an **identification of potential future green suppliers**, whose would most be best suited to supply green products in future similar events.

Potential usage of the output:

The output of this query may be utilized for a range of purposes, such as:

- 1) To **identify products** that not only meet quality standards (high rating scores) but also align with sustainability goals (lower average carbon footprints). This dual focus enhances the overall value of their procurement process in the future.
- 2) To **identify suppliers** of products that perform well in both sustainability and quality metrics.
- 3) To **provide historical data as a basis for benchmarking**, against which various other products may be evaluated.
- 4) To enable **continuous improvement** in the overall project's efforts towards green economy and sustainability, where targets can be made in real terms and acted upon.

Report 2-Top 3 Carbon Footprint during the event by details information within specific date period

Code:

```
SELECT DISTINCT P.productID, P.carbonFootprint, V.venueID, R.ratingDate
From ratingprodvenue RP
JOIN PRODUCT P ON RP.productID = P.productID
JOIN VENUE V ON RP.venueID = V.venueID
JOIN ratingsystem R ON RP.ratingID = R.ratingID
WHERE R.ratingDate BETWEEN '2032-08-04' AND '2032-08-07'
ORDER BY P.carbonFootprint DESC
LIMIT 3
```

Screenshot:

	productID	carbonFootprint	venueID	ratingDate
▶	P01	9.50	ST02	2032-08-07
	P30	9.30	ST02	2032-08-05
	P11	8.80	ST01	2032-08-06

Potential usage of the output:

Meaning:

This query lists the top 3 products with the highest carbon footprint. It is essential for identifying which products contribute most to environmental degradation. By reviewing this report, the client can focus on mitigating the use of high-emission products and work towards

more sustainable alternatives. This analysis directly contributes to the council's sustainability goals by allowing for targeted reductions in carbon emissions.

Environmental Monitoring: This report highlights the products contributing the most to carbon emissions, allowing organizations to focus on reducing their carbon footprint.

Strategic Decision-Making: This information assists in long-term strategic decisions, such as opting for more eco-friendly alternatives, adjusting procurement strategies, and meeting sustainability targets.

Venue Impact Assessment: By connecting product emissions to specific venues, this report helps organizations understand the environmental impact of different venues and how product usage may vary across them.

Report 3-Supplier Evaluation Standards

Code:

```
SELECT
s.supplierID,
s.supplierName,
SUM(p.qtyProduced) AS totalQtyProduced,
SUM(CASE WHEN p.reusabilityStatus = 'Recyclable' THEN p.qtyProduced ELSE 0 END) *
100 / SUM(p.qtyProduced) AS recyclingPercentage
FROM Product p
JOIN productsupplier ps ON p.productID = ps.productID
JOIN supplier s ON ps.supplierID = s.supplierID
WHERE p.qtyProduced > 0
GROUP BY s.supplierID, s.supplierName
ORDER BY recyclingPercentage DESC;
```

Screenshot:

	supplierID	supplierName	totalQtyProduced	recyclingPercentage
▶	SP02	Patrizia Kidsley	598	100.0000
	SP19	Noel Skillern	228	100.0000
	SP11	Maegan Robken	489	100.0000
	SP10	Christian Gosalvez	1078	58.6271
	SP03	Stephi Northrop	1369	56.9759
	SP04	Jethro Georgeou	1457	52.6424
	SP14	Parsifal Clayworth	850	0.0000
	SP12	Gilbertina Padefield	784	0.0000

Potential usage of the output:

- **Evaluate suppliers' environmental performance:** By comparing the total quantity produced and the recycling percentage, decision-makers can quickly identify the suppliers who meet sustainability standards. For example, suppliers like Patrizia Kidsley and Noel Skillern, who have a recycling rate of 100%, contribute significantly to waste reduction goals.
- **Optimize partnerships:** For future events, the city council can prioritize suppliers with high recycling rates, ensuring better resource reuse and reduced carbon emissions. This data serves as a reliable foundation for policy-making in environmental initiatives.

- **Promote improvements:** For suppliers with lower recycling rates (such as Jethro Georgeou with 52.64%), the city council can recommend improvements to help them increase their recycling performance or reconsider future partnerships.

Report 4 -Total carbon emission for each service

Code:

```
SELECT
    S.serviceName AS ServiceName,
    SUM(P.carbonFootprint) AS TotalCarbonEmission
FROM
    greenpackaging.service S
JOIN
    greenpackaging.servicevenue SV ON S.serviceID = SV.serviceID
JOIN
    greenpackaging.venue V ON SV.venueID = V.venueID
JOIN
    greenpackaging.ratingprodvenue RPV ON V.venueID = RPV.venueID
JOIN
    greenpackaging.product P ON RPV.productID = P.productID
GROUP BY
    S.serviceName
ORDER BY
    TotalCarbonEmission DESC;
```

Screenshot:

	ServiceName	TotalCarbonEmission
▶	Waste collection	169.60
	Sanitisation	169.60
	Hospitality	169.60
	Medical & health	169.60
	Field & equipment cleaning & maintena...	169.60

Note: In actual environment, the won't share the same result!

Potential usage of the output:

The Total carbon emission for each service report can be used in different areas and bring a lot of benefit. For instance,

- (1) Sustainability Monitoring: Help different events to track and reduce carbon emissions of different services.
- (2) CSR Reporting: Show the sustainability efforts in corporate social responsibility reports and help bring good brand emage.
- (3) Service Optimization: Identify and improve high-emission services to reduce costs and environmental impact, which helps link the service emissions to energy usage and operational costs for efficiency gains.
- (4) Marketing: Combine different events and service to promote low-carbon services and attract eco-conscious consumers.t.

Report 5 - Top 3 total carbon emission from sports

Code:

```
SELECT
    sp.sportID,
    s.sportName,
    SUM(r.totalCarbonFootprint) AS totalCarbonEmission
FROM sports s
JOIN sportproduct sp ON s.sportID = sp.sportID
JOIN product p ON sp.productID = p.productID
JOIN ratingprodvenue rpv ON p.productID = rpv.productID
JOIN ratingsystem r ON rpv.ratingID = r.ratingID
GROUP BY sp.sportID, s.sportName
ORDER BY totalCarbonEmission DESC
LIMIT 3;
```

Screenshot:

	sportID	sportName	totalCarbonEmission
▶	S16	Rowing	55
	S17	Sailing	47
	S15	Judo	47

Potential usage of the output:

This query displays 3 sports that have the highest carbon emission based on the products they use. This can be used to check whether there are any activities within sports events which carbon footprints may be reduced.

The total carbon emission for each sport report can be used in different areas and bring a lot of benefits. For instance:

- (1) It may help with **sustainability reporting** by identifying the activities within those sports that contribute to the high carbon emissions. Reduction of carbon footprint within these activities may also benefit other sports that have similar activities.
- (2) It may also be used for **CSR purposes**, especially enable suppliers to choose to endow supplierships to those sports that can be conducted with low carbon footprints. Such suppliers may use this strategy to enhance their corporate image.
- (3) It may also be used by **future** organizers as a **benchmark and target** to reduce carbon footprint in future sports events. Future activities can make use of this report to create new targets for lower carbon footprints.

Report 6 -Top 5 performance of product [reusable]

Code:

```
SELECT DISTINCT P.productID, P.reusabilityStatus, R.ratingScore
From ratingprodvenue RP
JOIN PRODUCT P ON RP.productID = P.productID
JOIN ratingsystem R ON RP.ratingID = R.ratingID
Where P.reusabilityStatus = 'Reusable' AND R.ratingScore = 15
ORDER BY P.productID
LIMIT 5
```

Screenshot:

	productID	reusabilityStatus	ratingScore
▶	P02	Reusable	15
	P06	Reusable	15
	P20	Reusable	15

Potential usage of the output:

This query identifies the top 5 reusable products with the highest performance rating (ratingScore = 15). By focusing on products that have both high reusability and top performance scores, the client can make informed decisions about which sustainable products to continue supporting or promote for future events.

Benefit:

This report is used to identifying the product that are reusable and here is the reasons

Sustainability tracking: It highlights the best-performing reusable products in terms of sustainability, which is important for organisations focused on reducing environmental impact.

Product Insights: By identifying top-performing reusable products, the report helps in decision-making regarding which products to prioritise or promote for future events or sales.

Operational Efficiency: It enables the organisation to focus on products that not only meet sustainability goals but also perform well, ensuring efficiency in both environmental and business operations.

Project planning: Iteration 1

Tasks	Team members	Hours allocated	Completion Date
Project proposal	Polly	0.5	1/9/2024
Narrative of the Functions-INPUT	Polly	0.25	4/9/2024
Narrative of the Functions-PROCESSING	Tara, Jiaying	1.5	4/9/2024
Narrative of the Functions-OUTPUT	Marco	1	4/9/2024
Entity-relationship diagrams	Tara	1.5	3/9/2024
Data dictionary-Sports	Tara	0.5	31/8/2024
Data dictionary-Venue	Marco	0.5	2/9/2024
Data dictionary-Product	Polly	0.5	1/9/2024
Data dictionary-supplier	Jiaying	0.5	2/9/2024
Data dictionary-Participant	Marco	0.5	2/9/2024
Data dictionary-Service	Jiaying	0.5	2/9/2024

Iteration 2

Tasks	Team members	Hours allocated	Completion date
Flowcharts	Polly	2	30 September
Input controls	All	5	26 September
Create and Insert SQL scripts	Tara	20	25 September
Explanation of security control	Marco Suen	6-8	30 September
Data cleaning and data quality maintenance plan	Jiaying Ying	8	30 September
Useful reports	All	10	2 October
PowerPoint slides preparation	All	1	13 October
Video recording	All	3	14 October
Video editing	Jiaying Ying	0.5	15 October

Final submission	Tara, Polly	2	17 October
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CONTINGENCY PLANNING

	Scenario Description	Plans on What to Do
1	When a group member is sick	The one with less job can take over his job for the moment / re-distribute work to other members. The sick member will take over more job after recovering.
2	When several group members have different ideas	Discuss the issue during the group meeting and vote to pick the best idea
3	When a member doesn't finish their tasks on time	Set up the deadline before submission and communicate. If the reason is acceptable, find out the solution together
4	When a member cannot be reached out / never shows up for meetings	Try to ask for his reason; if not acceptable, inform the course coordinator as soon as possible for a punishment.

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